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Research on C4.5 algorithm improvement strategy based on MapReduce

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Abstract

By analyzing the shortcomings of C4.5 decision tree algorithm, this paper combines MapReduce parallel model in Hadoop platform with C4.5 decision tree algorithm, so that C4.5 algorithm can be executed in parallel, so as to improve the efficiency of the algorithm. The improved C4.5 algorithm implementation process is given and the process is analyzed. Finally, the C4.5 algorithm improved experimental environment configuration is explained.

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1. Introduction

The focus of data mining work is the design and implementation of data mining algorithm. Therefore, improving the efficiency and accuracy of data mining algorithm has always been a hot issue. Parallelization of data mining algorithm is the best choice [1]. In order to improve the efficiency of algorithm execution, MapReduce parallel programming model is introduced into C4.5 algorithm.

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2. C4.5 analysis of advantages and disadvantages of decision tree algorithm

C4.5 decision tree algorithm was proposed by J.R. Quinlan in 1993. It is a decision tree generation algorithm based on ID3 algorithm. C4.5 algorithm overcomes the shortcomings of ID3 algorithm in application. In C4.5 algorithm, information gain rate is used as the basis of test attribute selection[2]. Although C4.5 algorithm has been widely used, there are still some shortcomings.

(1) C4.5 algorithm needs to select the maximum information gain rate as the split attribute when selecting the split attribute. Because the calculation process involves many logarithm operations, the library functions need to be called frequently which greatly increases the amount of calculation and reduces the efficiency of the algorithm.

(2) When the continuous attributes are discretized, the algorithm needs to calculate the information gain rate of all the segmentation points, which leads to a significant increase in the operation time when selecting attributes [3].

(3) In C4.5 algorithm, for classified attributes, the number of branches generated by each node is the number of attribute values. But there are many empty branches or some of them have no contribution to the classification. The existence of these branches reduces the availability of decision tree and leads to over fitting problem.

(4) C4.5 algorithm chooses the pessimistic pruning method. When there are many nodes in the decision tree, the calculation efficiency is low and the time is long which reduces the generation efficiency of the decision tree.

By analyzing the shortcomings of C4.5 algorithm, it is found that how to improve the execution efficiency of C4.5 algorithm is the core of C4.5 algorithm improvement and parallel execution is the best choice to improve the efficiency of the algorithm. Therefore, it is a beneficial exploration to combine MapReduce with C4.5 algorithm.

3. Analysis of MapReduce distributed parallel computing framework

MapReduce parallel computing framework is the core of Hadoop technology. It provides a computing mode for big data processing that can use the underlying distributed computing for parallel processing and provides a complete set of programming interface and execution environment for developers. MapReduce computing model is a standard functional programming mode[4]. The core of this computing mode is that one function can be passed as a parameter to another function. Such a function is usually called a higher-order function. Through the concatenation of multiple higher-order functions, the data computing process can be transformed into a series of function execution processes.

MapReduce divides the data processing task into map and reduce, which have their own processing functions map() and reduce(). The map() function is responsible for filtering and transforming the input in the format of < key, value > and outputting the intermediate data in the format of < key, value > [5]. Hadoop sets the values with the same key value together as the input of the reduce() function. The final calculation result is obtained through the centralized calculation of the reduce() function. The process is shown in Fig. 1:

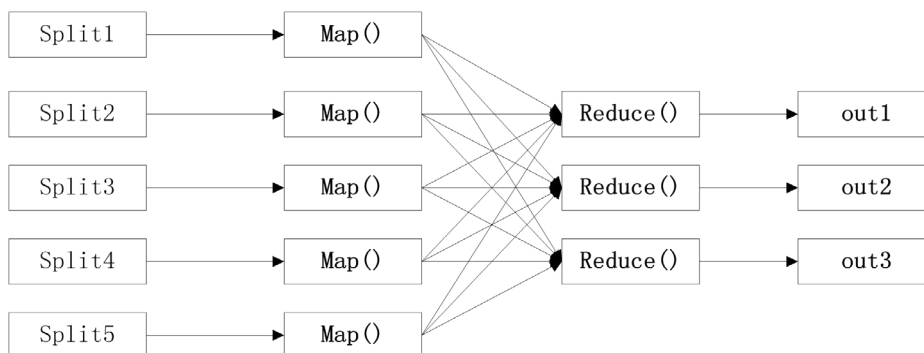


Fig. 1. MapReduce programming model

4. Improved design of C4.5 algorithm based on MapReduce

In order to improve the parallelism of decision tree algorithm, MapReduce is used to calculate the information gain rate of large-scale training samples and get the statistical information of each attribute. Then the information gain rate of attribute is calculated by using this information, so as to improve the degree of parallelism. In the process of building the decision tree with C4.5 algorithm, the nodes of each tree are determined by the best split point, that is the attribute with the highest information gain rate and the information gain not less than the average information gain condition [6]. The specific process is shown in Fig. 2:

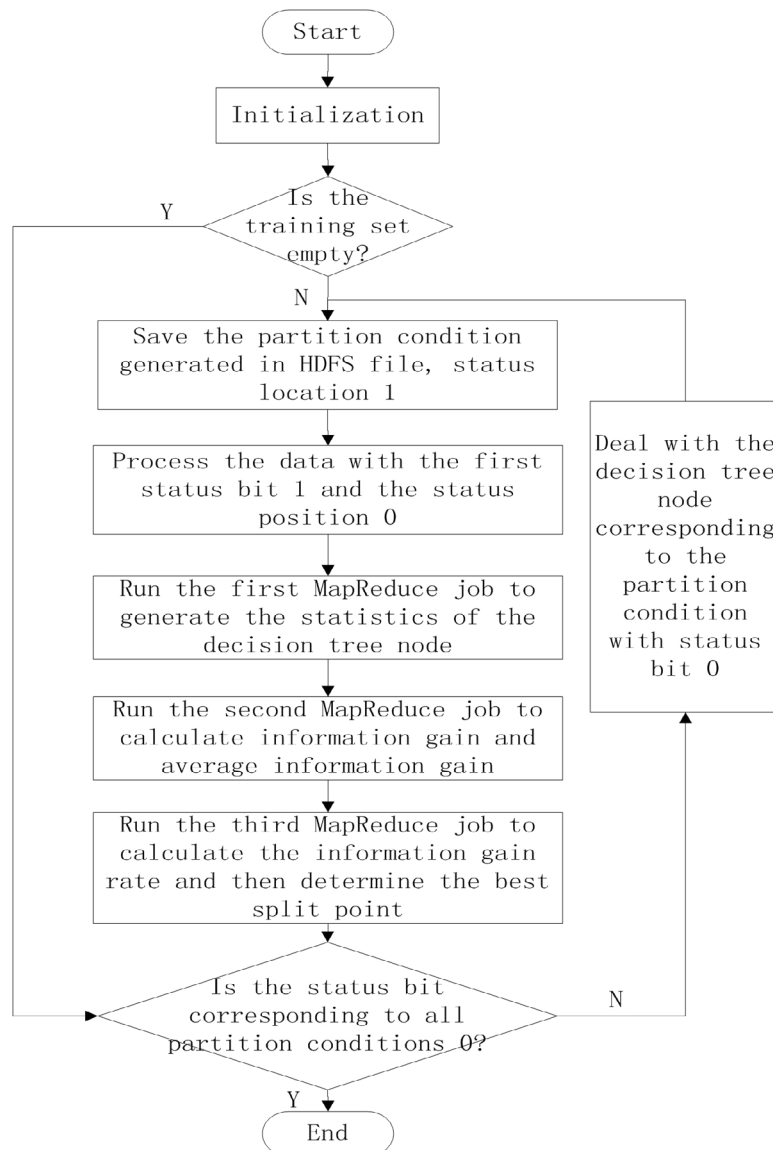


Fig. 2. decision tree algorithm flow chart of C4.5 parallelization

There are three jobs in the MapReduce process in this process. The first one is mainly responsible for counting the values of training samples on different attributes and class labels[7], the second one is mainly responsible for

calculating the information gain and average information gain when all attributes are taken different values, the third one is mainly responsible for calculating the information gain rate of all nodes and then remove the attribute whose information gain is lower than the average information gain[8] and select the attribute node with the largest information gain rate as the final partition node among the remaining attributes.

After obtaining the statistical information, MapReduce will calculate the information gain and information gain rate of each nodes according to the saved statistical information for each nodes in the current layer, it can find out the attributes that meet the best split point, and then divide each node[9]. For the newly divided sub nodes, it will check whether all their data have the same class label. If the newly divided sub nodes have the same class label, they will be written into the decision tree as leaf nodes. Otherwise, the data set represented by them will be added into the HDFS file and saved, and the state position 1 can be set to 0 for the state bit of the best processed partition point [10]. In the next call, only the best split point with flag 1 can be calculated. For the specific processing flow of each node is shown in Fig. 3

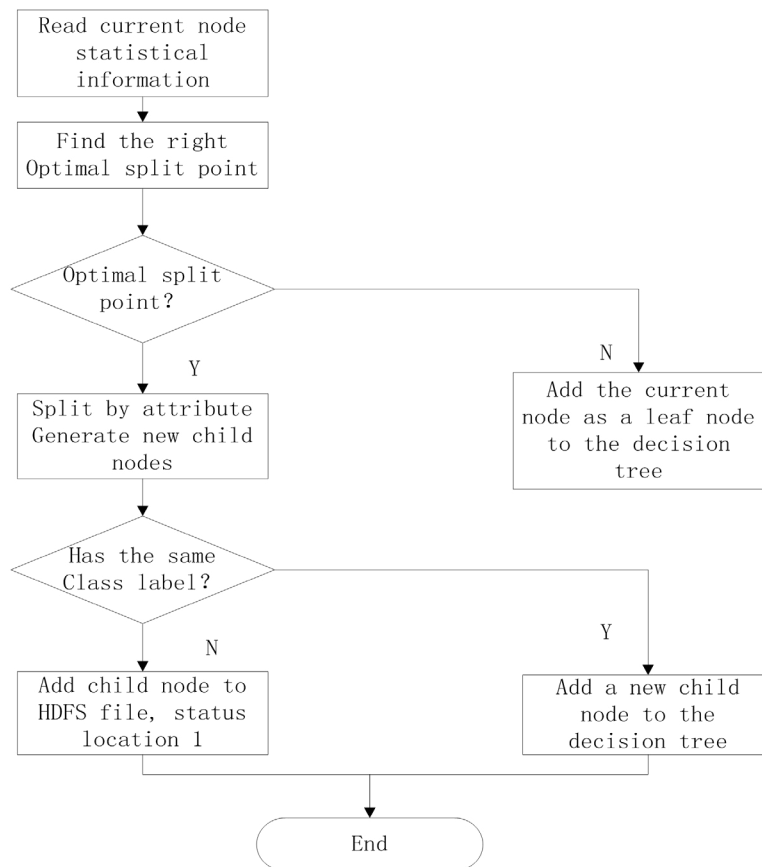


Fig. 3. MapReduce execution process

5. C4.5 improved algorithm experiment environment configuration based on MapReduce

1) Environment configuration

The parallel improvement of C4.5 algorithm is mainly through building Hadoop platform on Linux system to carry out the experiment of the algorithm. The whole experimental platform consists of five computers (including 1 host and 4 slaves). The specific configuration information is as follows[11]:

(1) Software configuration

Operating system: Linux, Hadoop: Hadoop 2.7.0.

(2) Hardware configuration

The configuration of five computers used in the experiment is:

CPU main frequency 2.8G, Memory 4G, Hard disk 500G.

2) Construction of Hadoop platform

The corresponding IP address and host name are stored in the respective files of each node. Since there is no DNS in this cluster, the corresponding IP address can be correctly resolved by modifying the `/etc/hosts` file. According to the specific situation, the distributed cluster has five nodes in total. The information of the five nodes in the cluster used in the experiment is shown in Table 1.

Table 1. Configuration information of experimental nodes

Host type	Host name	Operating system	IP address	Node type
Host 1	M1	Linux	192.168.0.51	Master
Slave1	M2	Linux	192.168.0.52	Slave
Slave2	M3	Linux	192.168.0.53	Slave
Slave3	M4	Linux	192.168.0.54	Slave
Slave4	M5	Linux	192.168.0.55	Slave

After the nodes are arranged, you need to complete the following preparatory tasks before installing Hadoop:

(1) Deploy the JDK environment on all machines. Because JDK is an essential part of Hadoop compilation and MapReduce running [12].

(2) Install and configure the keyless login environment. In Hadoop cluster, namenode and jobtracker can remotely control the services of all nodes in the slave list through SSH. However, it can only be accessed through authentication every time, which will reduce the working efficiency of the cluster. Therefore, in order to avoid mutual access between nodes, you need to manually type the key every time, so SSH is set to the form of login without key authentication.

(3) Install and configure Hadoop. Step1: on the primary node namenode, download the Hadoop installation file and extract it to the Hadoop user directory [13]. Step2: extract the obtained compression package on each node of the cluster to the directory with uniform name and configure the Hadoop home environment variable to the path name of the Hadoop folder obtained after decompression.

By selecting the appropriate parallel scheme, the improved algorithm can be successfully deployed to the Hadoop platform. By analyzing the flow of improved algorithm based on Hadoop, we can draw the following conclusions [14]:

(1) In the whole process of the algorithm, because the number of `map()` functions is related to the size of the read file, the parallelization process of map stage is very high.

(2) Because the C4.5 improved algorithm is only different in the processing order of nodes, the other steps are the same. According to the C4.5 algorithm parallel process, we can find that the series parallel C4.5 algorithm has the same output results. So the improved algorithm has no impact on the accuracy of the algorithm although it is deployed to the Hadoop platform.

6. Concluding remarks

This paper discusses the parallel strategy of C4.5 algorithm based on Hadoop which allocates the huge computation generated by processing massive data to the nodes of the cluster. The basic idea of improving C4.5 algorithm based on Hadoop is given. The distributed file storage and parallel computing are realized to improve the system performance.

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